Help the Doctor: How Much of Clinical Specialty Classification Is Reading the Text?

Group 33: Joel Graça (102256), Afonso Carvalho (116482) and Yann Vlk (120949).
Contribution: Joel Graça 33.3%, Afonso Carvalho 33.3%, Yann Vlk 33.3%

Instituto Superior Técnico, Universidade de Lisboa

1 Introduction

The task is to assign one of 12 medical specialties to a clinical note described by four fields: `description`, `sample_name`, `transcription` and `keywords`. When inspecting the data we found that two properties of the source corpus, MTSamples [Boyle, 2018], carry most of the label information: the `keywords` field usually starts with the name of the specialty, and many notes appear several times with a different specialty in each copy. This led to our research question:

RQ: *How much of the attainable performance comes from these dataset artifacts rather than from the clinical text, and which data augmentation strategy helps the part of the problem that does require reading the text?*

2 Data

2.1 Data Analysis

`train.csv` has 2,616 rows. In 3 of them, unescaped line breaks place text fragments in the label column; we discard them, keeping $N = 2{,}613$ notes. The test set has 398 notes. Median lengths are 405 words (transcription) and 16 words (description); 30 training notes (2 in test) have an empty transcription.

Imbalance. *Surgery* has 876 notes (33.5%), followed by *Cardiovascular-Pulmonary* (299) and *Orthopedic* (292); *Psychiatry-Psychology* (40) and *Dermatology* (25) are the smallest classes, a 35:1 ratio. The baseline (Table 1) has 0% F1 in 7 classes.

Observation 1: the label is written in `keywords`. In 2,172 notes (83.1%) the first keyword is a specialty name, e.g. "`neurology, temporal lobe, brain tumor, ...`". It coincides with the gold label in 2,171 of them (the exception is a note labelled *General Medicine* with keyword `obstetrics / gynecology`). The same pattern occurs in 320 of the 398 test notes. It is the website category of the note: metadata, not clinical content.

Observation 2: cross-listed duplicates with conflicting labels. 1,469 training rows (56%) share their transcription with at least one other row, forming 702 groups. In 701 of these groups the copies have *different* labels: the website lists an operative report under *Surgery* and also under the organ specialty. The most frequent pairs are *Orthopedic*/*Surgery* (118 groups), *Cardiovascular-Pulmonary*/*Surgery* (113) and *Gastroenterology*/*Surgery* (95). For instance, every note named "`Carpal Tunnel Release -` k" appears once as *Orthopedic* and once as *Surgery*. Both labels are clinically valid, so the task is partly multi-label presented as single-label: from the text alone, the two copies cannot be separated. 237 of the 398 test transcriptions also occur in `train.csv`.

2.2 Data Augmentation

Augmentation is applied only to the training part of each fold (never to evaluated notes). Every class with fewer than 150 training notes (*Obstetrics-Gynecology*, *Neurosurgery*, *Ophthalmology*, *Psychiatry-Psychology* and *Dermatology*) is filled up to 150 notes with synthetic copies of randomly sampled notes of that class. We compare four strategies: (i) **none**; (ii) **random oversampling (ROS)**: exact copies; (iii) **word dropout**: copies in which each word of `description` and `transcription` is removed with probability 0.15 (the "random deletion" operation of EDA [Wei and Zou, 2019]); (iv) **crossover**: a new transcription formed by the first half of one note and the second half of another note of the same class, which keeps the specialty vocabulary while creating unseen word combinations. We avoided synonym replacement: general thesauri do not cover clinical terms, and swapping an anatomy term changes the meaning.

3 Models

Pre-processing. Each field is lowercased; characters other than letters, digits, "-" and "/" are replaced by spaces. We do not remove stop words (negations such as "no" are relevant in clinical

notes) and do not stem, since stemming merges distinct terms (e.g. `neurology`/`neurologic`). The four fields are concatenated and represented with word uni- and bigram TF-IDF (sublinear term frequency, 25,000 most frequent n-grams, L2 norm).

Baseline. Multinomial Naive Bayes over TF-IDF of `description` with lowercasing and Porter stemming [Porter, 1980], as defined in the task.

Text-only classifiers. To measure what the text alone supports, the specialty term is removed from the head of `keywords` before training and prediction. We compare three linear classifiers with different inductive biases: (1) **Complement Naive Bayes** (CNB, $\alpha = 0.3$) [Rennie et al., 2003], a generative model designed for imbalanced text, with no class weighting; (2) **Logistic Regression** (LR, L2, $C = 1$), a discriminative probabilistic model; (3) **Linear SVM** (hinge loss, $C = 0.1$) [Joachims, 1998], a maximum-margin model. LR and SVM use `class_weight=balanced`, i.e. the loss of class c is multiplied by $N/(12\,N_c)$.

Duplicate rule (artifact 2). If the transcription of a note to classify is identical to one in the training data, the labels of those training copies are excluded and the classifier returns its highest-scoring *remaining* class. This follows from Observation 2: a copy almost never repeats the label of another copy.

Keyword rule (artifact 1). If the first keyword is a specialty name, it is returned as the prediction.

Hybrid system. Keyword rule when applicable; otherwise text-only classifier with the duplicate rule. This is our final system. For comparison we also train each classifier on the *full* fields (specialty term kept inside the bag of words).

4 Experimental Setup

All results are obtained with stratified 5-fold cross-validation (CV) on the 2,613 training notes (seed 42): the out-of-fold predictions (each note predicted once by a model that did not see it) are our own test set; the duplicate rule only looks up the training folds. `test_no_labels.csv` is used only to produce `results.txt`. We report accuracy, macro-F1 and minimum per-class F1 (the two task criteria are accuracy $> 49\%$ and minimum F1 $> 25\%$).

4.1 Parameters/Hyperparameters

C (inverse regularisation strength) and α (additive smoothing) were selected on the CV macro-F1 of the text-only setting among $C \in \{0.05, 0.1, 0.3, 0.5, 1, 3, 10\}$ and $\alpha \in \{0.1, 0.3, 1\}$. Augmentation target: 150 notes per class; dropout probability 0.15. Implementation: scikit-learn [Pedregosa et al., 2011] (LIBLINEAR [Fan et al., 2008] for SVM), pandas [McKinney, 2010], NumPy [Harris et al., 2020], NLTK [Bird et al., 2009] (Porter stemmer), Matplotlib [Hunter, 2007] and seaborn [Waskom, 2021]. The final model is trained on all 2,613 notes with the selected configuration (SVM + ROS).

Setting	Model	Acc	Macro-F1	Min-F1
Baseline	MNB (desc.)	37.5	11.4	0.0
Text-only	CNB	46.2	42.6	21.5
	LR	47.4	49.6	26.9
	SVM	47.2	50.6	29.7
Full fields	LR	52.4	54.0	28.6
	SVM	52.3	54.7	38.1
Text-only + dup. rule	LR	76.4	74.0	45.7
	SVM	77.2	74.9	50.0
Hybrid (keyword + dup. rule)	CNB	93.6	92.0	78.3
	LR	94.9	94.4	**81.9**
	SVM	**94.9**	**94.6**	81.6

Table 1: 5-fold CV results (%), all with ROS augmentation. "Full fields" keeps the specialty term inside `keywords`.

Subset of training notes	n	Text-only	Hybrid
Specialty keyword present	2,172	46.2	99.95
No keyword, duplicated	145	33.1	88.3
No keyword, unique	296	61.5	61.5

Table 2: Accuracy (%) of the SVM per subset (out-of-fold).

5 Results

Table 1 answers the first part of the RQ. Using all four fields more than doubles the baseline macro-F1, but the text-only classifiers stay below the 49% accuracy criterion. Keeping the specialty term in the concatenated bag of words adds only 5 points, because one token is diluted among 400 transcription words. The duplicate rule adds about 30 points and the keyword rule another 18. Table 2 breaks the hybrid SVM accuracy down by note type: the 296 notes without either artifact, the only ones where the decision depends entirely on the text, are classified with 61.5% accuracy.

Table 3 answers the second part. Augmentation matters only for CNB, the model without class weights: without it, *Dermatology* is never predicted (minimum F1 0%), and any of the three strategies raises macro-F1 by about 6 points. For LR and SVM, whose loss is already reweighted, no strategy improves macro-F1 by more than 0.8 points over no augmentation, and crossover lowers the minimum F1 of LR from 27.3% to 20.0%. Word dropout and crossover are not better than exact copies (ROS).

	CNB		LR		SVM	
Augmentation	M-F1	Min	M-F1	Min	M-F1	Min
None	35.3	0.0	**50.6**	**27.3**	49.8	29.2
ROS	**42.6**	**21.5**	49.6	26.9	**50.6**	**29.7**
Dropout	41.3	20.7	49.8	27.1	49.7	28.8
Crossover	41.1	20.0	48.5	20.0	50.3	29.4

Table 3: Effect of augmentation in the text-only setting: macro-F1 and minimum per-class F1 (%).

Class	P	R	F1	Class	P	R	F1
Cardio-Pulm	95.4	89.6	92.4	Ob-Gyn	96.2	92.6	94.3
Derm	100	92.0	95.8	Ophthalm	100	100	100
Gastro	100	88.5	93.9	Ortho	97.9	93.8	95.8
Gen Med	71.2	95.7	81.6	Psych	95.1	97.5	96.3
Neuro	94.0	94.5	94.2	Radiology	97.7	97.2	97.5
Neurosurg	93.8	96.2	94.9	Surgery	99.4	97.7	98.6

Table 4: Per-class precision, recall and F1 (%) of the final system (hybrid SVM + ROS), out-of-fold.

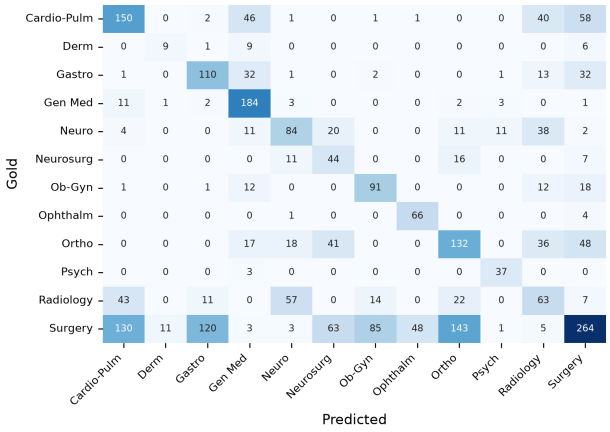


Figure 1: Confusion matrix of the text-only SVM (ROS), out-of-fold (rows: gold, columns: predicted).

Per class (Table 4), the hybrid SVM has F1 above 81% in every specialty; the lowest is *General Medicine*, whose precision (71.2%) is reduced by notes of other specialties predicted as *General Medicine*. Figure 1 shows the confusion matrix of the text-only SVM: 612 of the 876 *Surgery* notes are predicted as another specialty, mostly *Orthopedic* (143), *Cardiovascular-Pulmonary* (130), *Gastroenterology* (120) and *Obstetrics-Gynecology* (85), which are the most frequent duplicate pairs of Observation 2.

6 Discussion

Most common error. Of the 132 errors of the final system, 80 are notes of other specialties predicted as *General Medicine*, none of them with the specialty keyword. These are consultation notes with the SOAP structure ("chief complaint", "history of present illness", "review of systems"); this vocabulary is shared by all consultation notes and outweighs the few domain terms. Examples from our out-of-fold test set:

Ex. 1 (gold *Cardiovascular-Pulmonary*, predicted *General Medicine*): *"REASON FOR VISIT: I have been asked to see this 63-year-old man with a dilated cardiomyopathy [...] He presents with a chief complaint of heart failure."* The diagnosis is cardiological, but the referral template dominates. In TF-IDF, "cardiomyopathy" is one term among many *General Medicine* terms.

Ex. 2 (gold *Surgery*, predicted *Obstetrics-Gynecology*): *"Repeat low transverse cervical cesarean section with delivery of a viable female neonate, bilateral tubal ligation and partial salpingectomy."* We consider the prediction clinically correct; the gold label reflects the website category, not the content. The same applies to a *"dual-chamber pacemaker implantation"* labelled *Surgery* and predicted *Cardiovascular-Pulmonary*.

Ex. 3 (gold *Neurosurgery*, predicted *Psychiatry-Psychology*): *"first followup after shunt surgery [...] eight to ten years of progressive gait impairment, cognitive impairment, and decreased bladder control"*. The classifier ranks *Neurology* first, which is plausible for normal-pressure hydrocephalus; since the same note is labelled *Neurology* in the training folds, the duplicate rule excludes it, and the second choice is driven by "cognitive". This shows the limit of the rule: it removes the label already used but does not know which specialty the other copy belongs to.

Relation to the RQ. The answer is that most of the measurable performance comes from the artifacts: from the text alone, linear models reach about 47% accuracy on this split, and 61.5% on notes without duplicates, while the artifacts raise accuracy to 94.9%. Both rules use only given inputs and training labels, but neither reflects the ability to read clinical text [Kaufman et al., 2012]: on new notes a system would perform closer to the text-only figures. Augmentation is not the bottleneck: with a cost-sensitive loss, synthetic copies add nothing measurable; the limit is the label definition ("where the note was filed"), not the amount of minority-class data.

7 Future Work

Since the remaining errors are not caused by class frequency, we would (i) give more weight to the "diagnosis" and "procedure" sections, (ii) replace TF-IDF with a clinical pre-trained encoder, less sensitive to the consultation template, and (iii) keep all copies of a note in the same fold.

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Use of LLMs. Claude (Anthropic) was used as an assistant throughout the project: it proposed and wrote implementations of the pipeline and experiment scripts at our request, helped analyse the data, and drafted and revised the text of this paper for lexical and semantic correctness. The authors defined the task approach, chose which proposals to adopt, inspected the data and outputs, and verified every reported result by running the submitted code.